



PERFORMANCE AND EVALUATION OF COMPUTER SYSTEMS AND NETWORKS

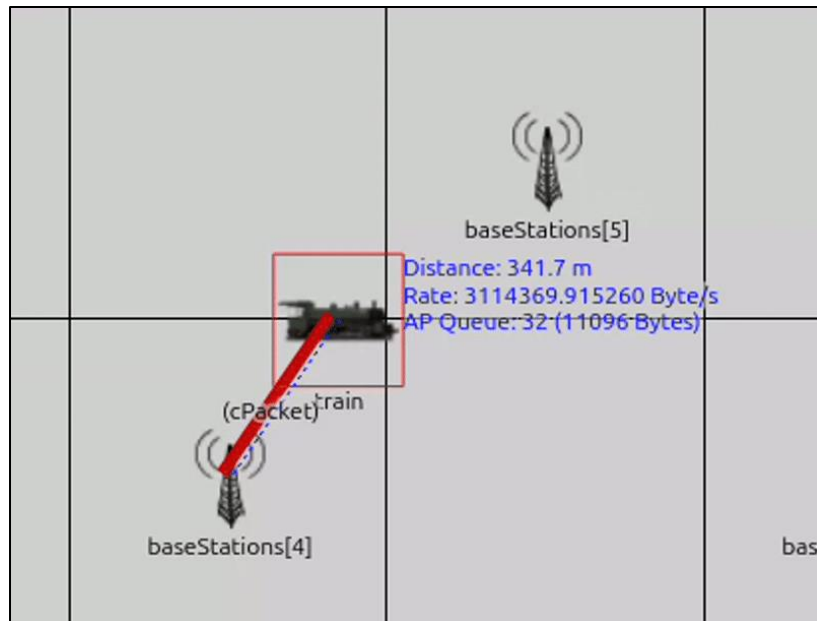
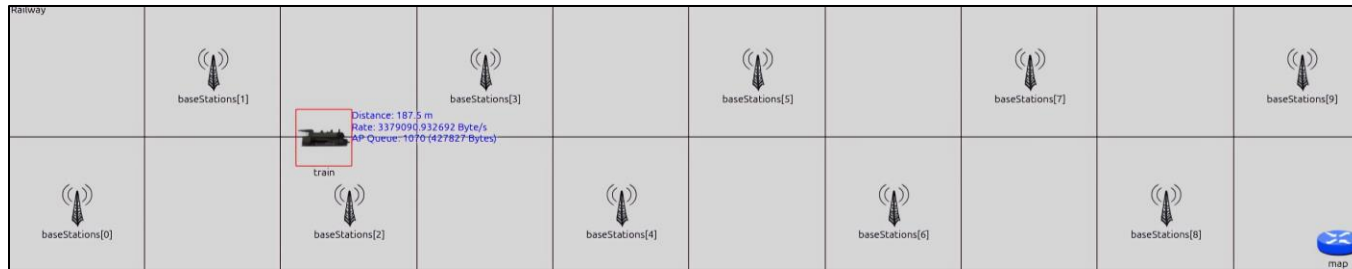
5G Connected High-Speed Train

Students:

Sebastiano Pala

Eleonora Sgorbini

Description of the system



Factor:

- **N:** Number of Passengers
- **M:** Number of Base Stations
- **H:** Mean Handover Delay (s)

Parameters:

- **S:** Mean packet size (Bytes)
- **T:** Mean generation time of the packets (s)
- **C:** Maximum transmission rate (Bytes/s)
- **V:** Train Speed (km/h)
- **L:** Railway length fixed to 10 km

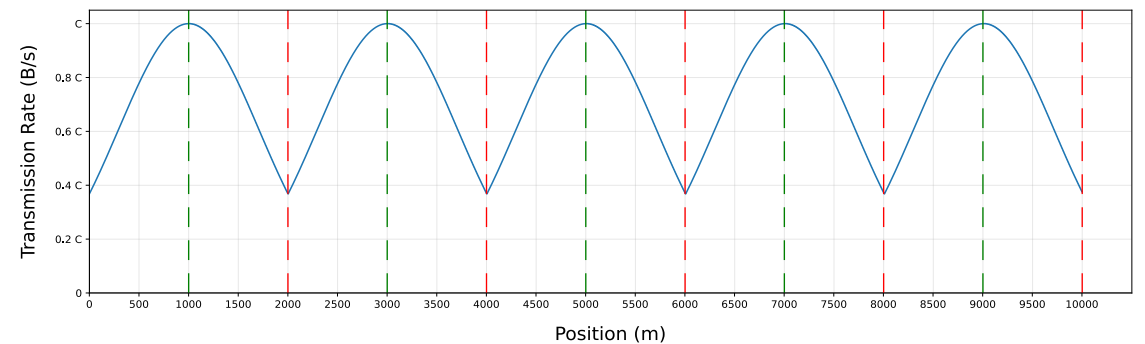
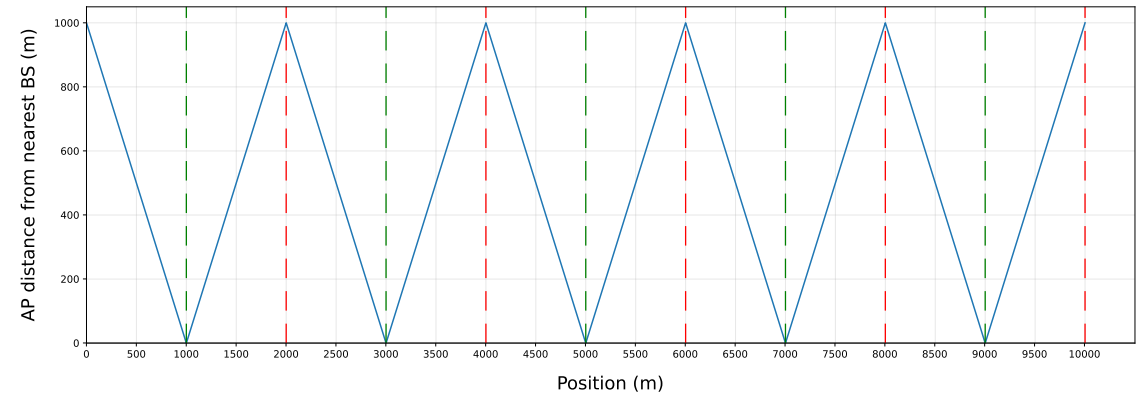
Performance Indexes:

- **Throughput** (Byte/s)
- **Response Time** (s)

Modelling

GENERAL ASSUMPTION

- ✓ The propagation delay between the Access Point and a Base Station is considered equal to zero
- ✓ The transmission time is calculated using the rate at the instant the transmission starts, and remains unchanged regardless of how the rate changes afterwards (due to the distance)
- ✓ The Access Point's queue is assumed to be infinite
- ✓ We have chosen to use a uniform random variable to represent the handover delay



Implementation

RAILWAY NETWORK

Modules

- **Base station:** simple module which models one of the entry points of the 5G radio network. All the base stations are evenly spaced along the railway
- **Mapper:** simple module which models an omniscient entity that knows all the base stations' positions
- **Train:** compound module which models the moving vehicle. This module contains:
 - **Passenger:** simple module which models one of the N passengers
 - **Access Point:** simple module which acts as a bridge between the passengers and the base station by receiving, storing and forwarding packets
 - **Simple Mobility:** simple module which updates the train's position
 - **Debug Module:** simple module which prints out values during the graphical simulation for debug purposes. It has no effect on the simulation

Verification

- **Deterministic Test**

Parameter	Value
Number of Passengers (N)	10
Number of Base Stations (M)	10
Handover Delay (H)	0 s
Train speed (V)	300 Km/h
Packet Size (S)	100 B
Generation Time (T)	1 s
Max Transmission Rate (C)	3.5 MB/s

- **Continuity Test**

	Simulation A	Simulation B	Simulation C
Mean Packet Size	300 B	325 B	350 B
Throughput	199502 B/s	216155 B/s	232808 B/s

- **Degeneracy Tests**

- $M = 0$ (No Base Stations)
- $V = 0$ (No Train speed)
- $N = 0$ (No Passengers)
- $C = 0$ (Max Rate = 0)
- $H = 0$ (No Handover)

- **Consistency Test**

	Simulation A	Simulation B
Passengers	20	40
Mean Generation	0.15 s	0.3 s

Throughput	53045 B/s	53038 B/s
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	Simulation B	Simulation C	Simulation D
Passengers	40	40	20
Mean Generation	0.3 s	0.15 s	0.3 s

Throughput	53038 B/s	106980 B/s	27148 B/s
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Calibration

Factor Calibration

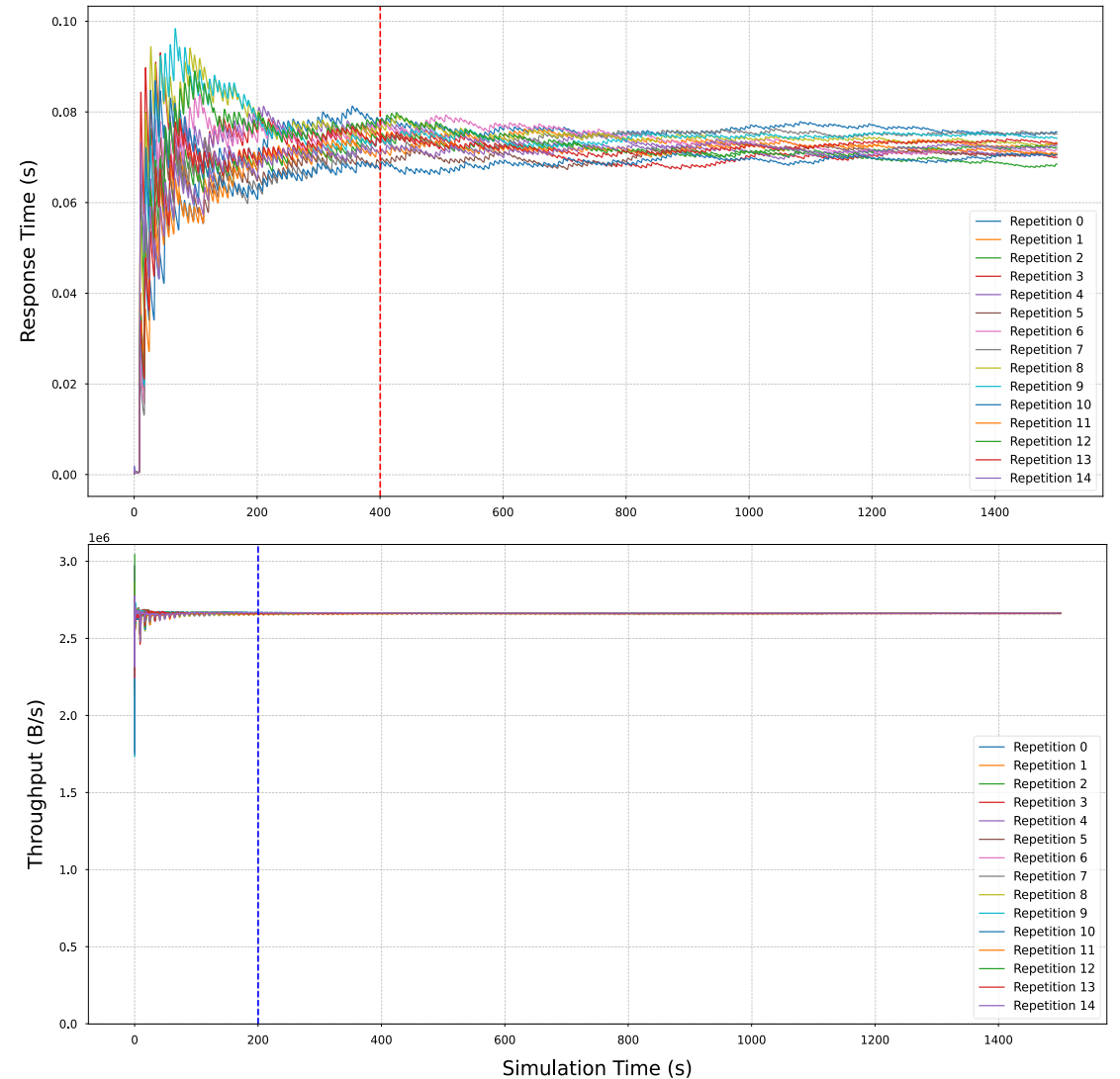
- Mean Packet size (S): 400 Bytes
- Mean Packet Generation Time (T): 0.03 s
 - Train speed (V): 300 Km/h
- Number of Passengers (N): {50, 100, 200, 300, 400}
- Number of Base stations (M): {4, 7, 10, 13, 16}
- Maximum Transmission Rate (C): 3.5 MB/s
- Mean Handover Delay (H): {0.3s, 0.7s, 1.1s}

Warmup period

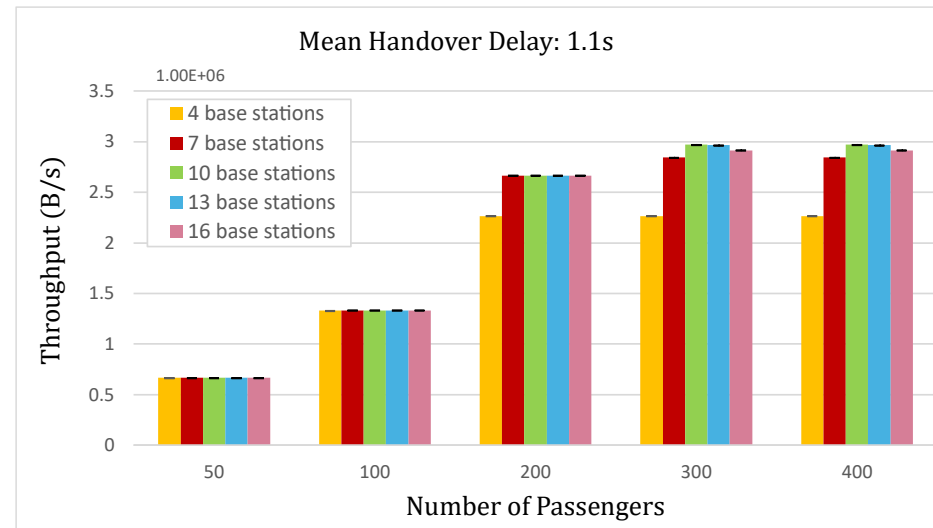
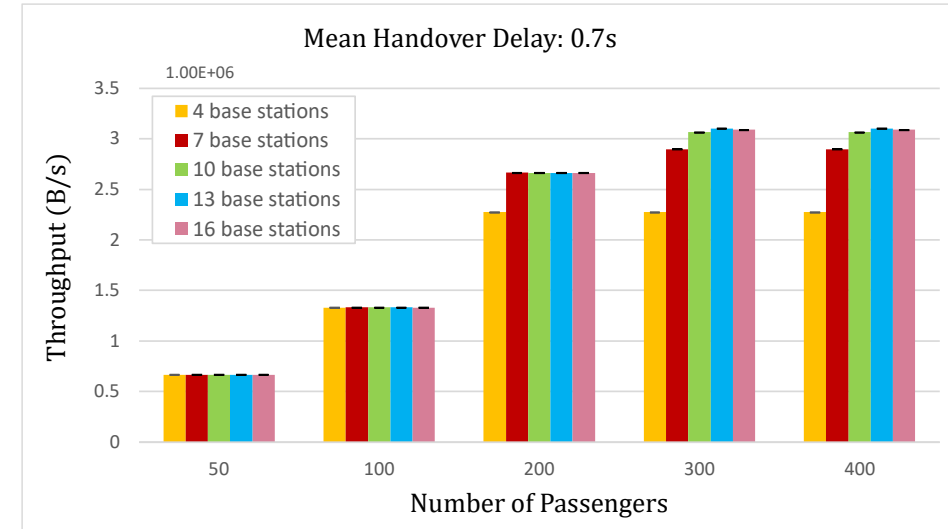
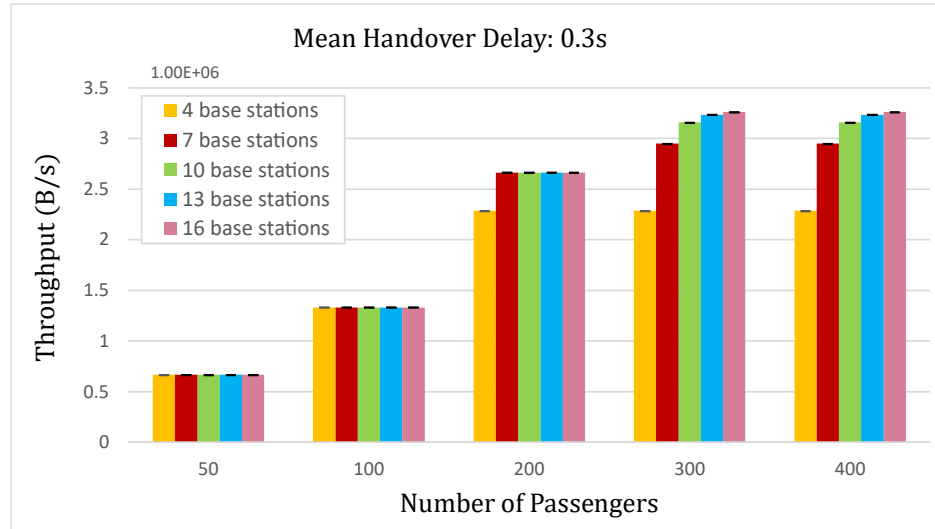
$$T_{warmup} = \max \{200s, 400s\} = 400s$$

Simulation Time

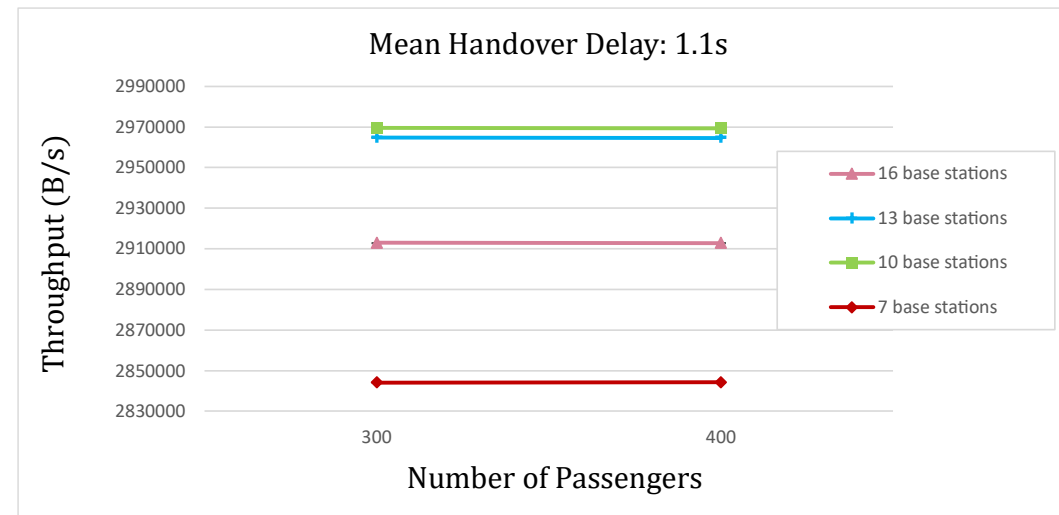
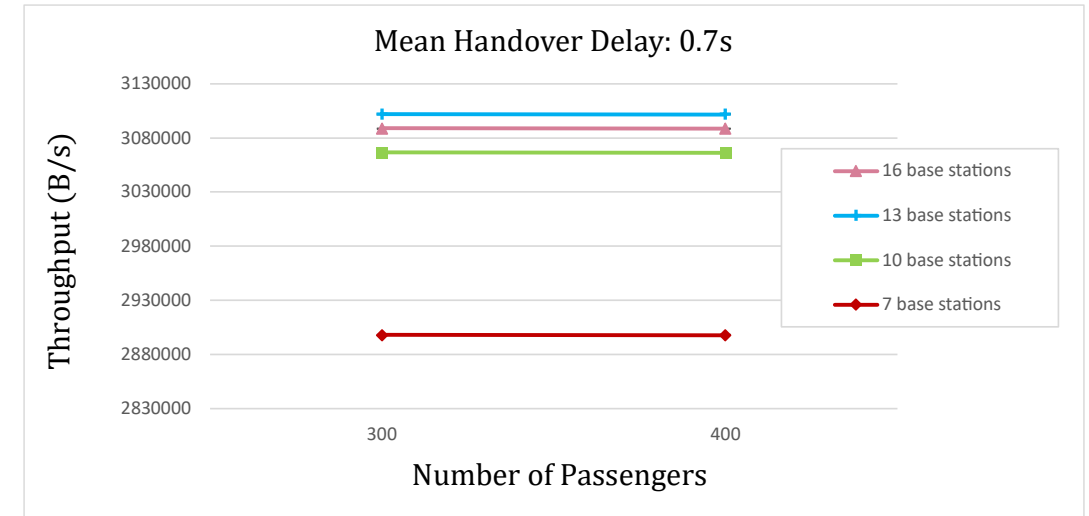
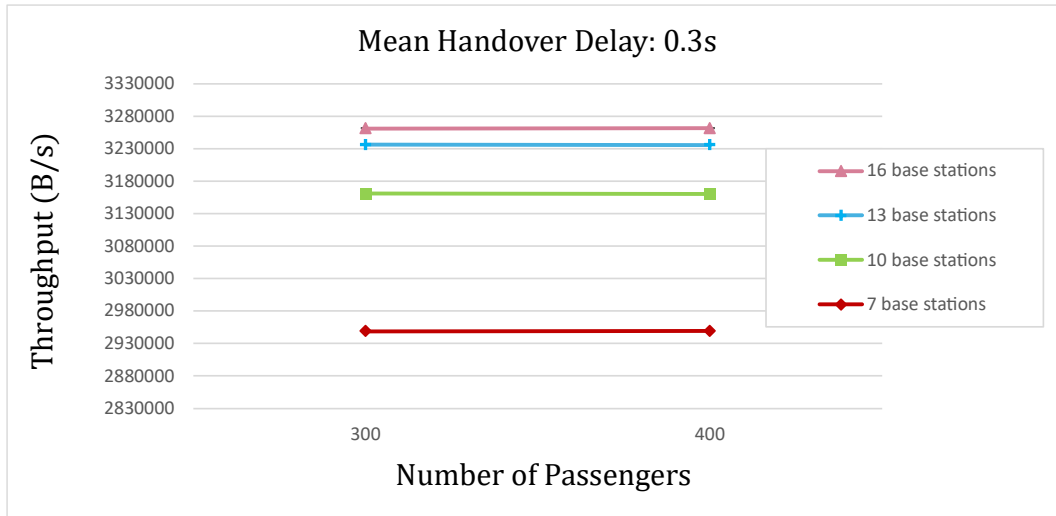
$$T_{simtime} = 1800s$$



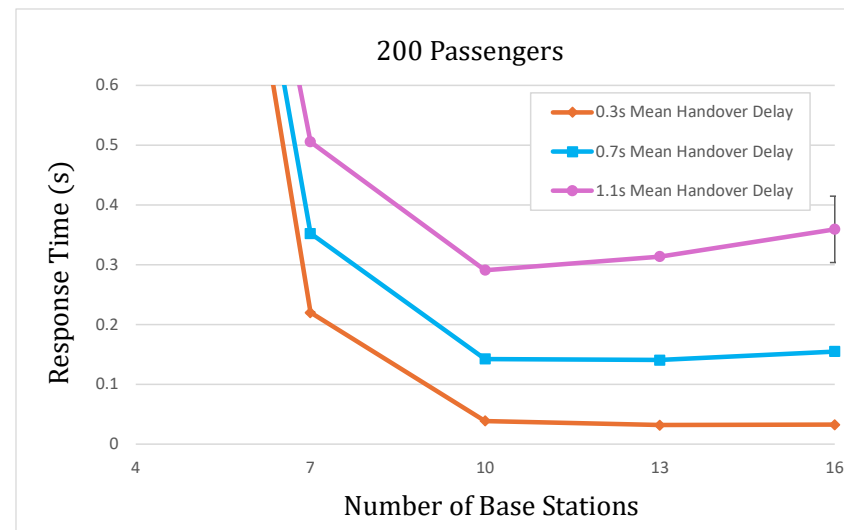
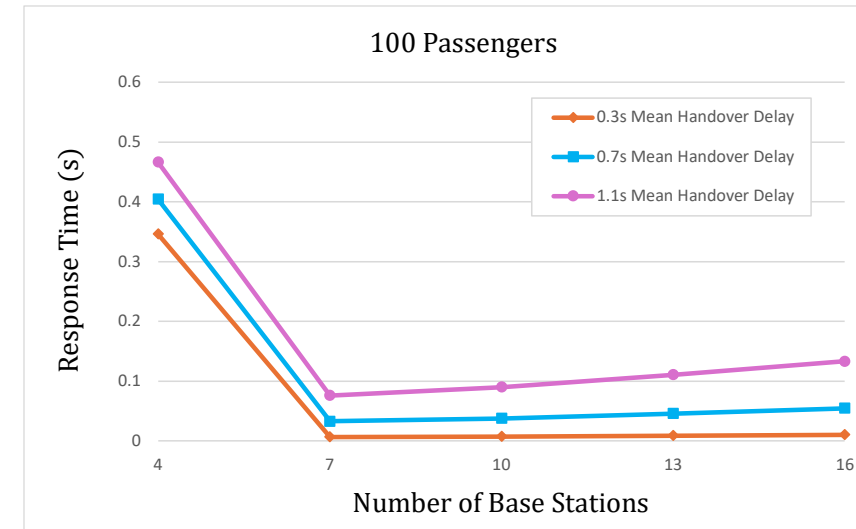
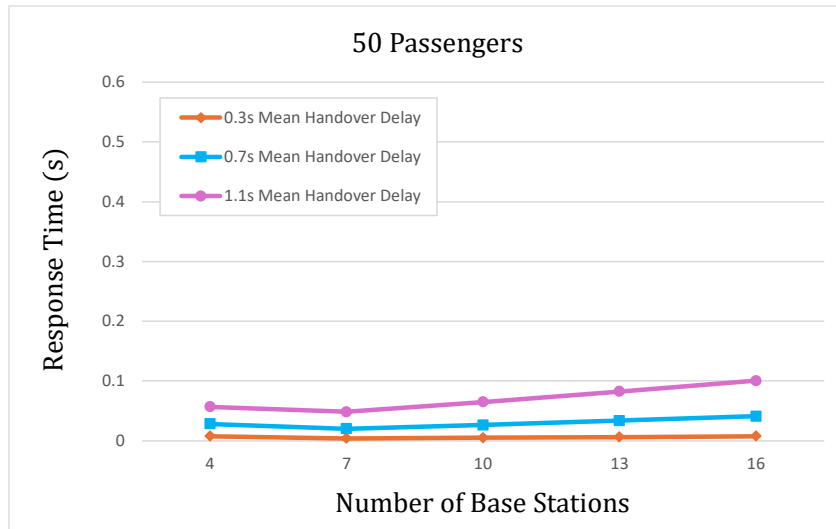
Experiments Design – Throughput Analysis



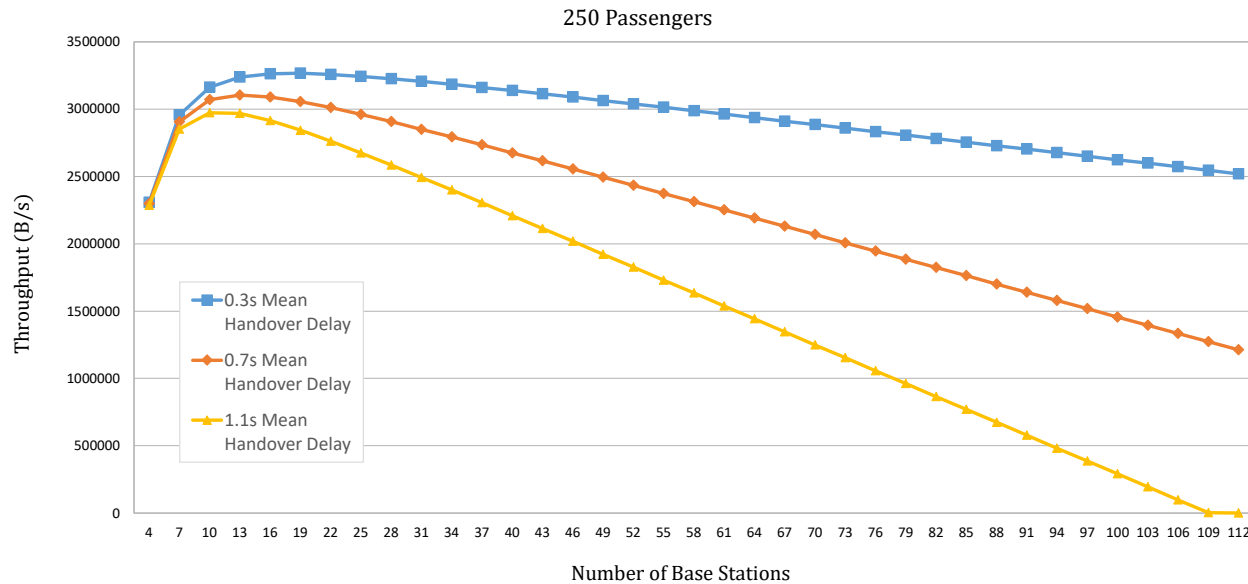
Experiments Design – High Load Scenario



Experiments Design – Response Time Analysis



Experiments Design – Bell Analysis

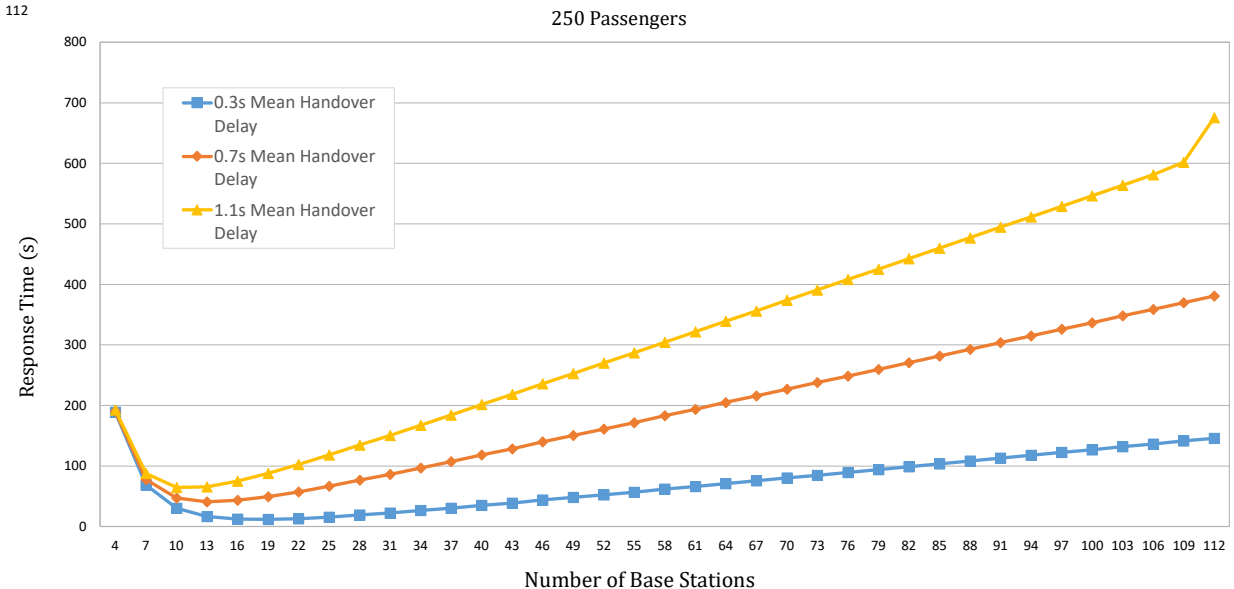


Interesting Observations

- The peaks shift to the left as the mean handover delay increases
- Non-linear behaviour up to the peak, followed by a linear trend
- With a huge number of base stations, transmission stops altogether

Factors Bell Analysis

Parameter	Value
Number of Passengers (N)	250
Number of Base Stations (M)	From 4 to 112 with step of 3
Mean Handover Delay (H)	{0.3, 0.7, 1.1} s



Conclusions

This study provided an analysis of 5G Connected High-Speed Train, highlighting the critical need to balance infrastructure sizing (the number of base stations) with the delay introduced by handover operations.

- **Low Load**
 - The system remains stable and throughput depends almost exclusively on the input rate
 - The throughput is a function of the number of passengers (N)
- **High Load**
 - The system reaches saturation, with throughput limited by the maximum transmission capacity
 - Increasing the number of base stations improves performance only up to an optimal point, after which diminishing returns are observed.
 - As the mean handover delay (H) increases, the peak performance shifts towards configurations with fewer base stations.

The simulations confirm that there is no single absolute optimal configuration. Maximizing throughput and minimizing response times requires a trade-off between the number of BSs and handover frequency.

Thank you
for your attention